ANGLO-CHINESE JUNIOR COLLEGE

JC2 Preliminary Examination

Higher 2

GEOGRAPHY **9173/02**

Paper 2 **28 August 2024**

3 hours

Answer Scheme

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Question 1

Question	Answer	Marks
1(a)	**Explain if the students' hypothesis and choice of sites are appropriate.** Award 1 mark for each explanation of appropriateness of sites relative to the hypothesis. Award a maximum of 1 additional mark for further development of each explanation. Possible responses include: *Hypothesis: Flood risk is higher in built-up areas relative to vegetated areas.* Choice of sites is appropriate in relation to the hypothesis on built-up area and vegetated areas. • The sites are in close proximity to both vegetated area which is in the Punggol Waterway Park, and in a drainage basin with a dense coverage of residential area with some commercial and residential land use. [1 mark] • While one of the site can be within the Park area, the second site should be placed in the residential area to collect data on infiltration and river variables data. [1 mark] This is due to the need to compare data collected within vegetated areas versus that of built-up area. Instead, both sites are within the Punggol Waterway Park. [1 additional mark] • In terms of size of potential area for research, it quite large, and could have potential for more sites, but based on the small group of investigators of six members, it is appropriate for them to have only two sample sites and placed both their sample sites within 600 metres of each other. [1 mark] This allows for quick access between the two places by walking instead of having to lug their fieldwork equipment using private or public transport. [1 additional mark]	4
Question	**Answer**	**Marks**
1(b)	**With reference to Resources 1 & 2, describe how the students could reduce the impacts of their investigation.** Award 1 mark for each description of how the impacts of their investigation can be reduced. Award a maximum of 1 additional mark for further development of each description. Possible responses: • Choice of day of investigation – change to a week day rather than a weekend. [1 mark] Will be in the way of park visitors. More during the weekend compared to the week day. [1 additional mark]	4

	• Should choose a different river system altogether. May not be able to retrieve the floating orange since the river is deep. [1 mark] The whole system of the waterway, the Serangoon River and the Punggol River is a reservoir for freshwater supply, and the floating orange would be a debris that contaminate the reservoir water. [1 additional mark] • The alternative is to have a long pole with a fish net at the base to retrieve the floating orange once it has gone through the finishing line of the velocity investigation. [1 additional mark] • To ensure that loosen soil of the infiltration site is restored. [1 mark] • To ensure that waste is bagged and placed in the appropriate waste collection bins. [1 mark] • To ensure that there is minimal noise generated so that park visitors can enjoy the peace and serenity park ecosystem service of the park. [1 mark]	
Question	**Answer**	**Marks**
1(c)	**Explain how the students can mitigate the potential risks of their fieldwork investigation at the sites shown in Resource 2.** Award 1 mark for each explanation of how to mitigate the potential risk of their fieldwork investigation. Award a maximum of 1 additional mark for further development of each description. Possible responses include: • Working in the locality of a river, not to go into the water at all. Good that they are obtaining data from the footbridge and not venturing into the river. [1 mark] This is especially since the depth of the Punggol Waterway channel at its deepest point along each of the cross section is around four metres. [1 additional mark] • While bending over to lower the string and plumb bob, to have some one hold on to the person or to wear a harness that is attached to a firm anchoring point to prevent falling over the railings of the footbridge. [1 mark] • Also to be aware of the location of the life-saving floatation device so that it can be deployed if any of the investigator accidentally falls into the river. [1 mark] • Should also be careful to wear footwear that provides a good grip since students will be working on the sloping banks of the river [1 mark]. This will help to minimise the potential of slipping down the slope into the river. [1 additional mark]	6

	• Water for the infiltration test should not be collected from the river channel, since there is a high potential for the student to slip and fall into the river that is fairly deep [1 mark]. Instead, student should collect the water from a nearby toilet facility [1 additional mark]. • Should have protective gear to protect against the sun since they are working several hours in the open are which do not appear to show any shading of any form be it artificial or natural [1 mark]. • Investigators should be briefed and are aware of the need to stop the field investigation once a thunderstorm occurs [1 mark]. This is to ensure that they will not be exposed to lightning strikes, slippery slopes and surfaces, and deep and fast flowing river waters [1 additional mark]	
Question	**Answer**	**Marks**
1(d)	**Explain how the reliability of the infiltration and river data shown in Resources 3 and 4 can be improved.** Award 1 mark for each explanation of how the reliability of their investigation can be improved. Award a maximum of 1 additional mark for further development of each explanation. Possible responses included: Infiltration data reliability can be improved upon in the following ways • At Site X – not to place the infiltrometer on a sloping surface. Easier to result in leakages from the bottom edge of the infiltrometer. This is evident in Resource 3 where the infiltration rate of at site x (16 mm/hr at the 90th minute) is significantly higher than the one at site Y with only 5 mm/hr at the 90th minute of the investigation [1 mark]. In addition, sloping grounds involves water that flows in two direction, vertically and also laterally. This increases the rate of transmission which in turn leads to a higher rate of infiltration. [1 additional mark] • The constant head method should be used instead of the falling head method. [1 mark] The constant head method allows for a more even hydraulic head of water that push water into the ground, and this is closer to the behaviour of rainfall infiltrating into the ground. [1 additional mark] • Change the infiltrometer used to one with a larger diameter, ie, it should be wider [1 mark] so that it is possible to view the water level close to the base of the infiltrometer especially when the falling head method of infiltration investigation is conducted [1 additional mark]. For the river data • Site Y cross section is at a river bend, and the river cross sectional shape is asymmetrical as seen in Resource 4. Site X shows a symmetrical cross section while Resource 2 shows that the site X is at the straight section of the Punggol Waterway channel [1 mark]. Since the idea is to see if different land use is the reason for variation in flood risk, this should be the only variable that is allowed to vary. Obtaining	6

	river variables when the channel characteristics are different makes the river data less valid [1 additional mark]. • Furthermore, generally river data should be obtained at straight sections of river instead of around a river bend. The floating orange may move with the swinging thalweg and the data would not be reflective of the overall average velocity of the river [1 additional mark]. • The orange that floats at the surface, provides river velocity readings of the river waters flowing at the surface which is the fastest. Thus it is suggested that more accurate river velocity data could be obtained a current meter is used instead of using the floating orange [1 mark]. Then river velocity readings can be taken at different parts across the river, and also at 0.2 and 0.6 depths to obtain more detailed readings to obtain the overall average readings of river velocity [1 additional mark].	
Question	**Answer**	**Marks**
1(e)	**Explain the data that the students should gather if they were to conduct a supplementary questionnaire survey on flood risk in the Punggol area.**	5

	Award 1 mark for each explanation of how the data can be gathered when conducting the supplementary questionnaire survey on flood risk in the Punggol area. Award a maximum of 1 additional mark for further development of each explanation. Possible responses included: • The number of times the resident has experience flooding in the Punggol area, or whether the resident or visitor to the park know of or have experienced a flood in the area of Punggol [1 mark]. • The number of time flood occurred in the area of Punggol [1 mark]. • Date and time when the flood occurred [1 mark] • The weather on the day of the flood [1 mark] • The nature of the flood in terms of areal extent, depth of water, and duration that the floodwaters remained [1 mark] • Any damage to properties and people [1 mark] • Do they know if there are any feature or structure in the area that may reduce the occurrence or the impacts of flooding in the area; ask if they can identify the feature or structure and its location 1 mark].	
Question	**Answer**	**Marks**
1(f)	**Explain the extent of the usefulness of the river data shown in Resource 4 in the investigation of flood risk.** Award 1 mark for each explanation of how the usefulness of the river data in the investigation of flood risk. Award a maximum of 1 additional mark for further development of each explanation. Possible responses included: Usefulness is from the perspective of whether it contributes to the knowledge of the extent of flood risk from the hazard point of view • Knowing the cross-sectional shape and the depth of water allows for the data of the wetted perimeter to be derived. The higher the wetted	5

	perimeter, the more frictional drag and hence slower the flow of water [1 mark]. Water flowing slowly indicates the stormwaters will take a longer time to flow out of the location leading to high flood risk [1 additional mark] • The hydraulic radius can be derived by using the formula of channel cross sectional area divided by the wetted perimeter. The hydraulic radius also provides a gauge to the efficiency of the channel shape in conveying the stormwaters away [1 mark]. A higher hydraulic radius indicates a more efficient shape with lower frictional drag allowing stormwaters to move out of an area faster. Hence the channel is more efficient at site X with a hydraulic radius of 3.22 as opposed to the one at site Y with 2.9 [1 additional mark]. • The limitation is that there are only two point data obtained. [1 mark] • Another limitation is that the relief of the surrounding area beyond the river can be relevant in evaluating the flood risk. Locations with higher relief leads to lower flood risk [1 mark]. • Gathering data on whether there are flood defences in the Punggol area would supplement the river data [mark]. • Knowing the geology of the area would be helpful as well since flood risk is significantly higher with impermeable rocks and soils [1 mark].	
Question	**Answer**	**Marks**
1(g)	**The students concluded that vegetated areas show lower flood risk compared to built-up areas. With reference to Resources 1, 2, 3 and 4, evaluate the validity of the students' conclusion.** Possible approaches: Students' conclusion that the vegetated areas show lower flood risk compared to built-up area is NOT valid for the following reasons: • The key reason is the locations where the data was gathered. **River data** is collected **only in the location near to vegetation**. No river data were collected in locations where the river or channel passes through residential area. • Infiltration data which provides information on the amount of rainfall that can infiltrate into the ground and the amount of overland flow that may be generated when rainfall does not infiltrate and hence may lead to higher risk of flooding is helpful in determining flood risk. There are only **two locations where the data was collected**, and both are in location with vegetation. There should be infiltration data gathered from the built-up areas as well. Collect data not only from vegetated surface, but concrete surfaces. Have more infiltration sites – go beyond the boundary of the park, into the residential area; sites can be over concrete or tarmac surfaces typical of built-up areas. • In addition, the **infiltration data** may not be valid since at one site, it was on flat ground (site Y), while at site X, it was on sloping ground. This included an additional variable that can affect infiltration besides that of land use.	10

	• The **timing of the data collection** would also render the data not valid. Both the river data and the infiltration investigation were conducted at two different times of the day. At site X, it was in the morning, where temperature would be lower, less water would have evaporated from the ground, antecedent soil moisture would have been higher, while the investigation at site Y was conducted after lunch in the afternoon. The higher temperature, the higher amount of water evaporated, and accordingly the antecedent moisture would be lower. Again, this would have introduced an additional variable of antecedent moisture into the infiltration data instead of just the land use variable alone. • **Accuracy of the infiltration data** is also compromised as there is a high chance that the readings may have been erroneous since the infiltrometer used as narrow and tall. Furthermore, the **falling head** method of infiltration data was used. It would be very difficult to obtain accurate reading when the water level is close to the base of infiltrometer which is only 10 cm wide, but 30 cm tall.	

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the distribution of the Small Island Developing States (SIDS). Resource 6 shows the temperature anomaly versus sea level rise. Resource 7 shows the impacts of climate change on aquatic and terrestrial ecosystems. Resource 8 shows an infographic about the impacts of climate change on SIDS.

(a) Using Resource 5, describe the locations of the SIDS in the world. [4]

Award 1 mark for each description of the locations of SIDS, up to 4 marks. Award 1 additional mark for further development where applicable.

Possible responses include:

- SIDS are located predominantly in the tropics.
- They are found in the Caribbean, the Pacific Ocean, and the Indian Ocean.
- A significant number of SIDS are scattered across the Pacific Ocean, such as Fiji and Tonga.
- Caribbean SIDS like Jamaica and the Bahamas are concentrated in the western Atlantic.
- Most are located in the ocean with the exception of Guyana which is part of South America.

(b) Explain the role of human activities in accelerating warming through positive feedbacks. [4]

Award 1 mark for identifying a human activity that accelerates warming up to 4 marks. Award 1 additional mark for further development or linkage to positive feedbacks.

Possible responses include:

- Deforestation: Large-scale deforestation removes trees and vegetation, which play a critical role in absorbing CO_2 through photosynthesis [1 mark]. In addition to increasing atmospheric CO_2, deforestation exposes large areas of soil, particularly in tropical regions. The exposed soil has a lower albedo (reflects less solar radiation) than the original forest cover, meaning it absorbs more heat, leading to an increase in local and global temperatures. The warmer temperatures can lead to more tree loss and further deforestation, perpetuating the warming cycle [1 additional mark].
- Melting Ice and Snow Cover: Human-induced global warming causes the melting of polar ice caps and glaciers [1 mark]. Ice and snow have high albedo, reflecting a large portion of solar radiation back into space. As they melt, darker land or ocean surfaces are exposed, which absorb more solar radiation. This increased heat absorption accelerates global warming, leading to more ice melt and further increasing temperatures in a positive feedback loop [1 additional mark].

(c) Using Resource 6, describe the relationship between temperature anomaly and sea level rise. [4]

Award 1 mark for describing the general relationship between temperature anomaly and sea level rise to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

There is a general positive correlation between temperature anomaly and sea level rise; as temperature anomalies increase, sea levels also rise.
According to Resource 6, in 2010, with a 0.5°C temperature anomaly, sea levels had risen by approximately 0.2 meters, and by 2020, a 1°C anomaly corresponded to a rise of around 0.3 meters.
As the anomaly increased to 1.5°C by 2050, sea levels are projected to rise by 0.4 meters, and at 2°C, the sea level is expected to rise by 0.5 meters by 2100.
However, in 1995, when sea levels had risen significantly by 0.2 meters, but the temperature anomaly remained relatively low at 0.3°C.

(d) Using Resources 6 and 7, explain the impacts of climate change on aquatic and terrestrial ecosystems. [6]

Award 1 mark for each explanation on the impacts of climate change on aquatic and terrestrial ecosystems to a maximum of 6 marks. Award up to 1 additional mark for further development or elaboration of each point.

Aquatic Ecosystems: Rising sea temperatures and increased ocean acidity, as shown in Resource 6 and 7, lead to coral bleaching. Coral reefs are highly sensitive to even small changes in temperature, and as sea temperatures increase, large-scale coral bleaching events occur. This weakens reef ecosystems, reducing biodiversity and disrupting food chains. For example, coral bleaching in the Great Barrier Reef has led to the collapse of fish populations that depend on healthy coral for shelter and food.

Terrestrial Ecosystems: Resource 7 indicates that rising temperatures and changing precipitation patterns result in the shifting of biomes. For example, forests in temperate regions may transition to savannas or grasslands due to increased drought and heat stress. This causes species migration, where certain plant and animal species relocate to higher altitudes or latitudes in search of cooler conditions, leading to ecosystem shifts and possible species extinctions if they cannot adapt.

Water Cycle Disruption: Climate change is disrupting the global water cycle, as shown in Resource 7. Increased evaporation rates due to warmer temperatures reduce freshwater availability in many terrestrial ecosystems, leading to droughts that stress plant and animal populations. This is particularly severe in arid regions where water is already scarce, exacerbating desertification and reducing biodiversity.

(e) Explain why there are variations in negative impacts of contemporary climate change due to uneven changes in temperature and precipitation across places. [6]

Award 1 mark for each explanation of how varying temperature and precipitation changes cause uneven impacts of climate change, up to 6 marks. Award an additional mark for further development of each explanation where applicable.

Possible responses include:

Geographical Location: Regions closer to the poles, like the Arctic, are warming faster than the global average, resulting in more severe impacts such as permafrost thawing and sea ice loss, while tropical regions experience slower temperature changes.

Precipitation Patterns: Areas that already receive high rainfall, such as tropical rainforests, may experience intensified monsoons and flooding, while arid regions like the Sahel suffer from prolonged droughts, worsening desertification.

Vulnerability of Ecosystems: Some ecosystems, like coral reefs, are more sensitive to even small temperature increases, whereas others, such as boreal forests, may be more resilient to initial changes but face long-term threats from fire and pest outbreaks.

(f) Using Resource 8, explain how countries can respond to contemporary climate change. [6]

Award 1 mark for each strategy explained to a maximum of 6 marks.
Award 1 additional mark for further development.

Beach Nourishment: Beach nourishment or protection of beach. This involves artificially replenishing beaches with sand to combat coastal erosion and rising sea levels. By expanding and maintaining beaches, these countries protect coastal areas from storm surges and create a buffer zone against flooding and prevents beach degradation.

Construction of Artificial Islands: This allows countries like the Maldives to adapt to rising sea levels by creating elevated land where essential infrastructure can be built. These islands, designed to withstand the effects of climate change, provide a long-term solution to displacement caused by rising sea levels.

Building Sea Walls: SIDS can also sea walls as a key defense against rising sea levels and coastal flooding. Sea walls act as physical barriers to protect coastal cities and critical infrastructure from storm surges. These walls are essential in preserving both livelihoods and the infrastructure that supports the local economy

Buying Land in Other Countries: In extreme cases, some SIDS are exploring the option of purchasing land in other countries as a last-resort adaptation strategy. Resource 8 mentions that countries like Kiribati have initiated plans to buy land abroad as a contingency for relocating their populations if sea levels rise to unmanageable levels.

Section C

Answer EITHER question 3 OR question 4.

3 Evaluate the strategies in creating liveable cities for different social groups. [20]

Suggested Approach for the question: "Evaluate the strategies in creating liveable cities for different social groups."

Candidates could approach the question by discussing the various strategies aimed at improving liveability for different social groups, including women, the elderly, and other vulnerable populations. These strategies could be classified into top-down approaches (government-led initiatives, policies, infrastructure) and bottom-up approaches (community-led projects, grassroots initiatives). Candidates could evaluate the effectiveness of these strategies in addressing the specific needs of different groups and ensuring inclusivity in urban planning. Additionally, the evaluation should consider the spatial and temporal scales at which these strategies are implemented and their impacts on social equity, access, and well-being.

Indicative Content:
Supporting the Strategies:

- **Top-down Strategies for the Elderly**: Governments have implemented various policies aimed at improving urban liveability for the elderly, including age-friendly infrastructure and housing options. For example, Singapore's Enhancement for Active Seniors (EASE) program provides grants for elderly-friendly features such as ramps and safety measures in homes, improving mobility and independence. Such policies show that infrastructure can be adapted to meet the physical needs of elderly individuals, helping them maintain autonomy and feel integrated into society(h2 l7 women)(L6 Needs of the elderly).
- **Gender-responsive Urban Planning**: Incorporating gender mainstreaming into urban planning addresses the unique needs of women, particularly in terms of safety, mobility, and access to public spaces. Strategies such as Women's Safety Audits in cities like Delhi, India, have helped identify areas where women feel unsafe, leading to urban design changes that prioritize safety and accessibility. These gender-sensitive initiatives ensure that women's needs are not overlooked, creating more inclusive cities(h2 l7 women)(h2 l7 women).
- **Affordable Housing for Low-income Families**: Strategies to create affordable housing, such as building mixed-income housing developments or offering rental subsidies, directly impact low-income groups' access to decent living conditions. In many cities, providing affordable housing options within accessible locations helps address socio-economic disparities, contributing to overall liveability(L6 Needs of the elderly).

Counter Argument:

- **Unequal Access and Implementation**: Although strategies like age-friendly infrastructure or gender-responsive urban planning are beneficial, not all social groups have equal access to these services. For example, while retirement villages may cater to high-income elderly individuals, low-income seniors may not afford such facilities and remain reliant on family care or face isolation. This highlights that many strategies primarily benefit wealthier segments of the population, limiting their inclusiveness.
- **Challenges in Bottom-up Approaches**: Grassroots and community-led initiatives, while valuable for fostering inclusivity, often face challenges in

scalability and long-term sustainability. Local projects such as community gardens or women's support groups are effective in addressing immediate needs but may lack the resources or institutional support to have a widespread impact on liveability across cities.

- **Overemphasis on Physical Infrastructure**: Some strategies focus heavily on physical infrastructure, such as building elder-friendly parks or accessible public transport, but may neglect the social aspects of liveability, such as promoting social inclusion and mental well-being. For instance, the creation of ramps and benches is helpful, but if there is no community engagement, the elderly may still feel socially isolated.

Evaluation: While strategies to create liveable cities for different social groups have shown success in improving urban infrastructure and inclusivity, their effectiveness varies depending on the group's specific needs and the context in which these strategies are implemented. Top-down approaches provide essential large-scale infrastructure improvements, but they may fail to reach all social groups equally, particularly in low-income populations. Bottom-up approaches, on the other hand, offer community-based solutions but often struggle with scalability. To create truly liveable cities, a combination of both top-down and bottom-up approaches, addressing both physical and social aspects, is necessary to ensure that all groups, particularly vulnerable populations, benefit from urban development.

4 'Mitigation strategies are the best response to contemporary climate change.'

Evaluate this statement. [20]

Suggested Approach

Candidates could approach the question by assessing the effectiveness of mitigation strategies in addressing the root causes of climate change, particularly focusing on reducing greenhouse gas emissions and preventing further climate impacts. They could then compare this with adaptation strategies, which aim to cope with the impacts of climate change rather than address its causes. Candidates could analyze the contexts in which mitigation strategies are more suitable, such as in developed nations with higher carbon footprints, and where adaptation strategies might be more necessary, such as in vulnerable developing regions. The evaluation could also include considerations of the long-term vs. short-term effectiveness of mitigation and adaptation strategies, as well as economic, political, and social feasibility.

Indicative Content:
Supporting the Statement:

- **Reduction of Emissions**: Mitigation strategies, such as transitioning to renewable energy sources, carbon capture, and reforestation, address the root cause of climate change by reducing greenhouse gas emissions. For instance, the Paris Agreement's focus on limiting global temperature rise through emission reduction targets highlights the centrality of mitigation in combating climate change. These strategies are essential in preventing future climate impacts by stabilizing the global climate system.
- **Long-term Benefits**: Mitigation strategies provide long-term solutions to climate change. While adaptation focuses on managing the symptoms (such as building sea walls or developing drought-resistant crops), mitigation directly addresses the cause, which is essential for preventing the escalation of climate-related disasters in the future. For example, widespread adoption of electric vehicles and

renewable energy infrastructure reduces dependence on fossil fuels, ensuring a more sustainable future.

- **Cost-effectiveness in the Long Run**: Mitigation strategies, although initially costly, can be more economically viable in the long term by avoiding the significant future costs associated with adapting to extreme weather events, rising sea levels, and other impacts of unchecked climate change. Investments in green technology and carbon-neutral industries may also spur innovation and create new economic opportunities, especially in developed countries with the resources to lead such transitions.

Counter Argument:

- **Urgent Need for Adaptation in Vulnerable Regions**: In regions already severely affected by climate change, particularly Small Island Developing States (SIDS) and low-lying coastal areas, adaptation strategies are more immediately necessary. Rising sea levels and extreme weather events require countries like the Maldives to focus on adaptation efforts, such as constructing sea walls and developing climate-resilient infrastructure. For these regions, mitigation alone cannot prevent the immediate threats they face, highlighting the importance of adaptation in the short term.
- **Feasibility of Mitigation in Developing Countries**: Many developing countries may not have the resources or infrastructure to implement large-scale mitigation strategies, such as transitioning to renewable energy. These nations often rely on fossil fuels for economic growth and poverty reduction, making it challenging to prioritize mitigation over adaptation. In countries like India or Nigeria, adaptation strategies, such as improving water management systems or enhancing agricultural resilience, are more pragmatic responses to the immediate impacts of climate change.
- **Time Lag of Mitigation Effects**: Even if mitigation strategies are implemented globally, the effects of emission reductions will take time to materialize. Climate change impacts already in motion, such as rising temperatures and more frequent extreme weather events, necessitate adaptation strategies to cope with these challenges in the short to medium term. This demonstrates that while mitigation is crucial, it must be complemented by adaptation strategies to manage the current and near-future impacts of climate change.

Evaluation: Mitigation strategies are essential for addressing the root causes of climate change and preventing further environmental degradation. However, their long-term benefits may not address the immediate challenges faced by vulnerable regions, particularly those already experiencing severe climate impacts. In such cases, adaptation strategies are necessary to protect communities and infrastructure in the short term. Thus, while mitigation remains the best response for long-term climate stability, a combination of both mitigation and adaptation strategies is needed to effectively respond to contemporary climate change, depending on the specific regional and temporal context.